



AI-Ready Data Products

Starburst Office Hours

An interactive webinar
with Lester Martin from DevRel

Connection before content

Lester Martin – <https://linktr.ee/lestermartin>

- Developer Relations @ Starburst
 - Blogging & forums
 - Webinars & videos
 - User groups & events
 - Training & tutorials
- 30+ years of technology experience
 - Started journey on TRS-80 Model III
 - Played most roles, but a programmer at my core
 - ½ career in OLTP and ½ in data analytics
 - Decade+ of “big data” experience to include
 - Trino/Starburst, Hadoop, Hive, Spark
 - NiFi, Kafka, Storm, Flink
 - HBase, MongoDB

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What is Context?

Business Descriptions

In business terms, what does the data mean?

Business Rules

What company specific definitions, conventions, and policies govern how data is interpreted

Encoding Quirks

What non-obvious patterns or gotchas exist in the data?

Synonyms

How else can people refer to something?

Null Semantics

Are nulls allowed, and what do they represent?

Derived Metrics

How are named business metrics computed from columns?

Relationships & Joins

How do columns connect to others and what SQL joins it?

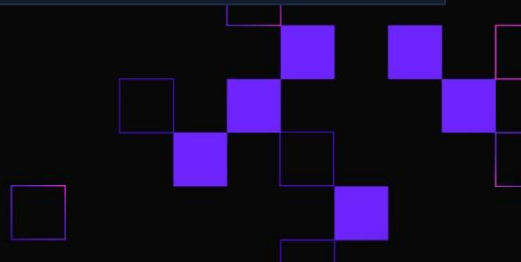
Value Enumeration

What are valid or representative values?

Row Representation

What does each row record represent?

...and more...



What Context Delivers: From Plausible to Truthful Answers

Accuracy

Right interpretations, joins, filters, and logic

Consistency

Same question results in same answer

Safety

Prevents incorrect data or inappropriate info from being used

AI makes context non-optional

LLMs can generate plausible but incorrect answers aka “hallucinations”

Strong data and business context are one of the most critical elements for successful enterprises to reduce hallucinations



Data products are our
context layer

Add and edit metadata for richer context

Capture context at every level

Add schema, table, and column descriptions via UI or API

Propagate context from catalog to data products

Inherit metadata from catalog into data products automatically

The screenshot displays a data catalog interface. On the left, a sidebar shows a hierarchical catalog structure with a search bar containing 'monica'. The main area is titled 'high_priority_vw' and shows the 'Columns' tab. A table lists the columns with their names, descriptions, types, and nullability. To the right, a 'Business rules for San Francisco Public Safety Data' panel provides context about the data product, including its purpose and governance standards.

high_priority_vw

Columns

Name	Description	Type	Nullable	Default
custkey	Unique identifier for the customer associated with the order.	bigint	Yes	null
orderdate	The date on which the order was placed.	date	Yes	null
orderkey	Unique identifier for the order record.	bigint	Yes	null
orderpriority	Priority classification of the order (e.g., 1-URGENT, 2-HIGH, 3-MEDIUM).	varchar	Yes	null
totalprice	Total monetary value of the order, expressed as a decimal.	double	Yes	null

Business rules for San Francisco Public Safety Data

This data product provides read-only access to curated and governed data for analysis and exploration.

The data within this product:

- Is sourced from approved systems and follows defined governance standards.
- Reflects the most recent available refresh based on the product's update schedule.
- May include transformations, aggregations, or filters applied for consistency and usability.

Explore new options to create/maintain Data Products

Data Products via Platform

Create and maintain Data Products inside Starburst

Create data product

Define data product

Define your data product by adding a title and description, selecting a catalog, and creating a schema. Title and description is the information users will see as they search for data products.

General information

Data product title *

Salesforce Report Data Product

Schema name: salesforce_report_data_product

Catalog *

iceberg_glue_mv

Schema name *

salesforce_report_data_product

Domain *

finance

Private

Data product summary *

This data product explains the Salesforce Report information available with the data product.

50 characters remaining

Data product description *

This data product will be active in March of 2026, and will look back from March 2025 until April of 2024. The point of this data product is a historical load of all Salesforce account data within the specific date range.

Cancel Save as draft Save and continue

1 Define data product

Title

Salesforce Report Data Product

Catalog

iceberg_glue_mv

Schema name

salesforce_report_data_product

Domain

finance

Summary

This data product explains the Salesforce Report information available with the data product.

Description

This data product will be active in March of 2026, and will look back from March 2025 until April of 2024. The point of this data product is a historical load of all Salesforce account data within the specific date range.

2 Define datasets

3 Data product details

Build and manage data products via existing visual UI

Data Products as Code

New method for managing Data Products outside of Starburst UI

```
# Column Metadata Template
# All fields except column_name and data_type (sourced from metastore).
# Remove fields that are not applicable for a given column.

# --- Business Context ---
business_name: 'Human-friendly display name for the column'
business_description: 'Plain language description of what the column means and how it is used.'
business_synonyms:
  - 'synonym one'
  - 'synonym two'
  - 'synonym three'

# --- Nullability ---
nullables: true # Omit if false
null_count: 0 # Number of null rows observed in sample
nullability_semantics: 'Explanation of why the column is null and what null means in business terms.'

# --- Cardinality & Values ---
cardinality: 'low' # low or high
distinct_count: 0 # Number of distinct non-null values observed
exhaustive_values: # Use for low-cardinality columns (<= ~15 distinct values)
  - 'value one'
  - 'value two'
sample_values: # Use for high-cardinality columns instead of exhaustive_values
  - 'example one'
  - 'example two'
  - 'example three'
value_range: # Use for numeric or date columns where range is meaningful
  min: 'minimum_value'
  max: 'maximum_value'

# --- Casting ---
cast_required: true # Omit if false
cast_notes: 'Exact instructions on how and when to cast, including example SQL.'

# --- Aggregation ---
aggregation_rules: 'Instructions on how this column must or must not be aggregated. E.g. always use AVG not SUM, never use COUNT(DISTINCT)'

# --- Relationships ---
fk_role: 'FK => other_table.other_column' # or 'PK of table_name' for primary keys
fk_group: 'group_name' # e.g. customer_links, payment_links, employee_links, lookup_links
join_snippet: 'LEFT JOIN other_table AS alias ON this_table.column = other_table.column'

# --- Quirks & Grouping ---
encoding_quirks: 'Any non-obvious encoding patterns, gotchas, cross-column dependencies, or SQL pitfalls.'
column_group: 'logical_group_name' # e.g. identifier, demographics, address, credit_card, mortgage, auto_loan, payment, geography, census
```

Maintain a reproducible YAML configuration of data products, allowing users to create, update, and overwrite via code



Demo